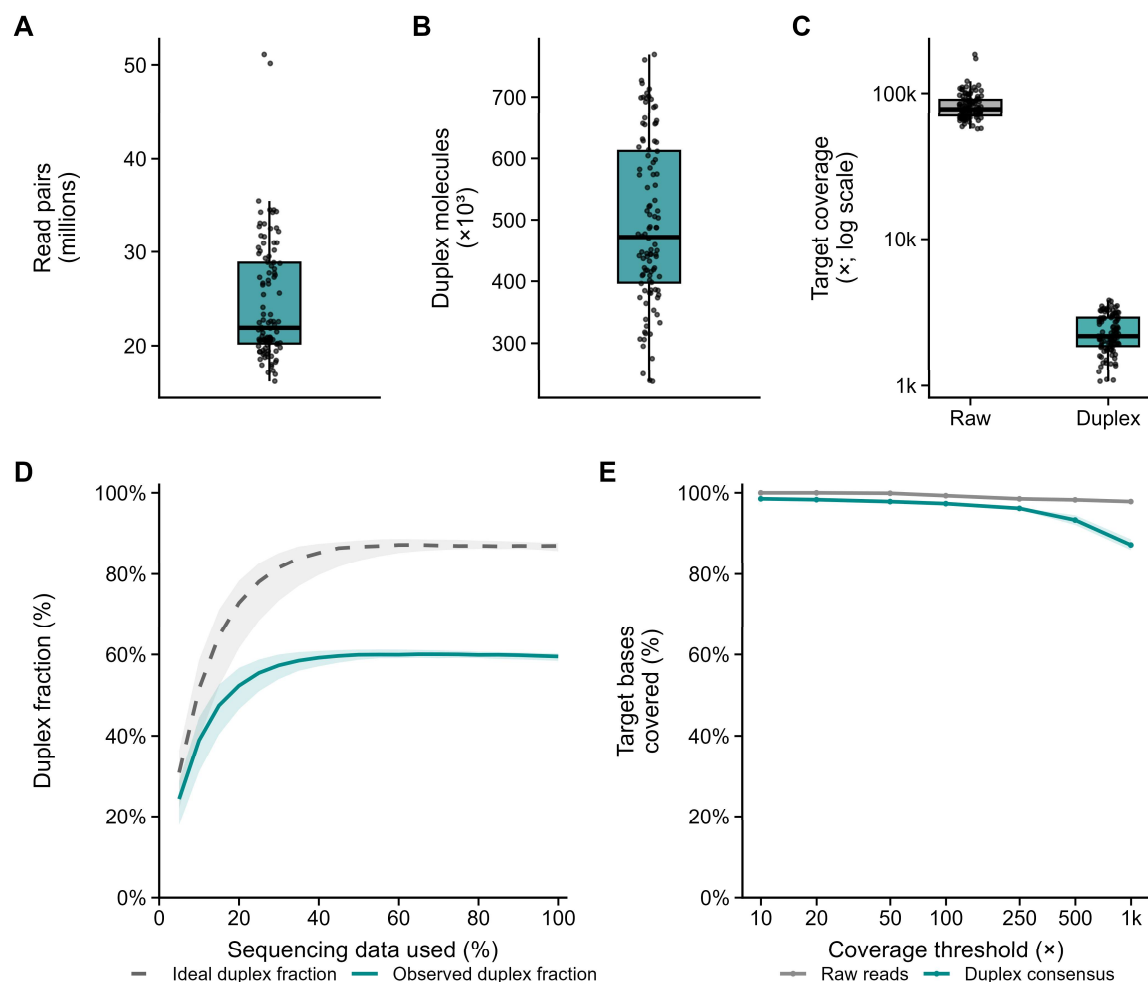
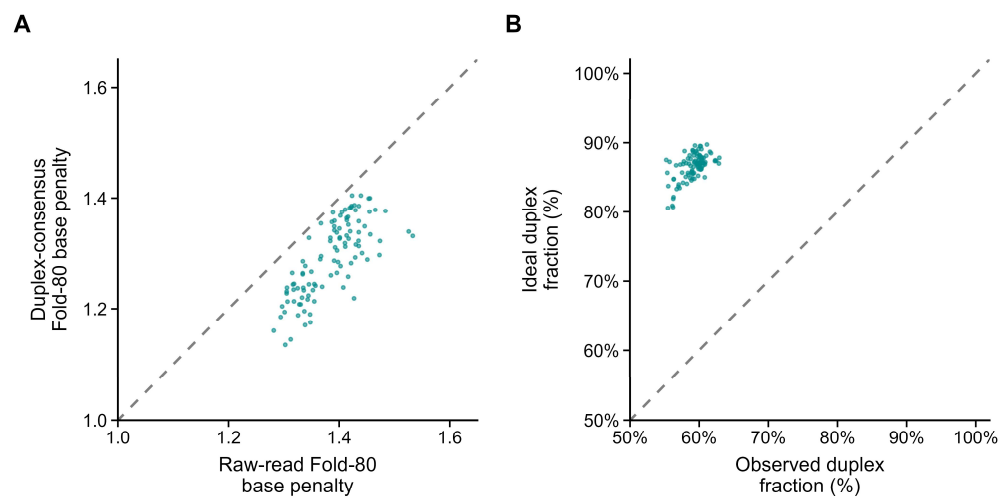


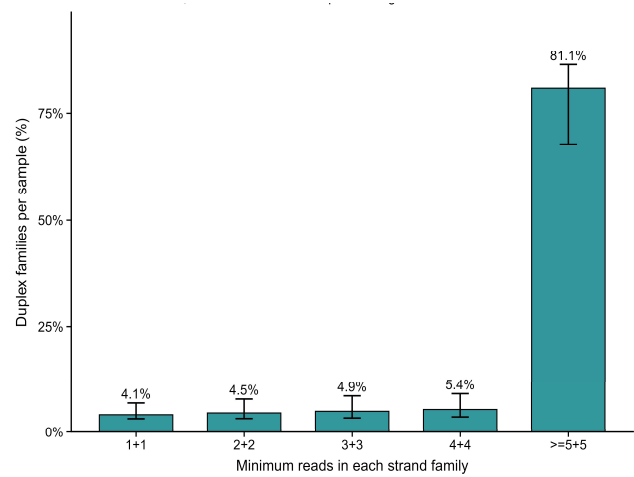
Supplementary Figure S1. Sequencing and duplex-consensus performance. (A) Sequencing read pairs per sample. (B) Independent duplex molecules per sample. (C) Mean target coverage before and after duplex-consensus generation. (D) Observed and ideal duplex fractions across increasing proportions of sequencing data. (E) Percentage of targeted bases covered across increasing coverage thresholds for raw and duplex-consensus reads. Boxes show medians and IQRs, and points show individual samples. Lines and shaded regions show medians and IQRs, respectively. Analyses include 104 patient plasma samples.



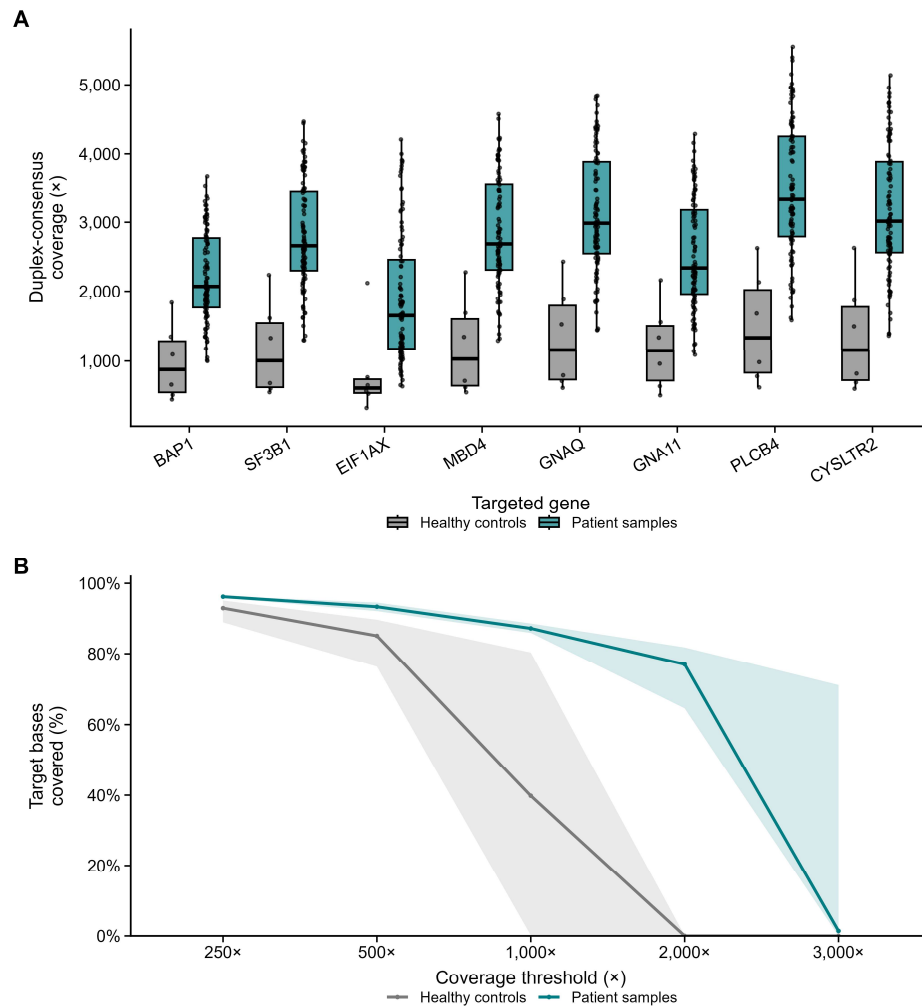
Supplementary Figure S2. Coverage uniformity and duplex fraction. (A) Raw-read versus duplex-consensus Fold-80 base penalty. Lower values indicate more uniform coverage. (B) Observed versus ideal duplex fraction at full sequencing depth. Dashed lines represent equality between measurements, and each point represents one of 104 patient plasma samples.



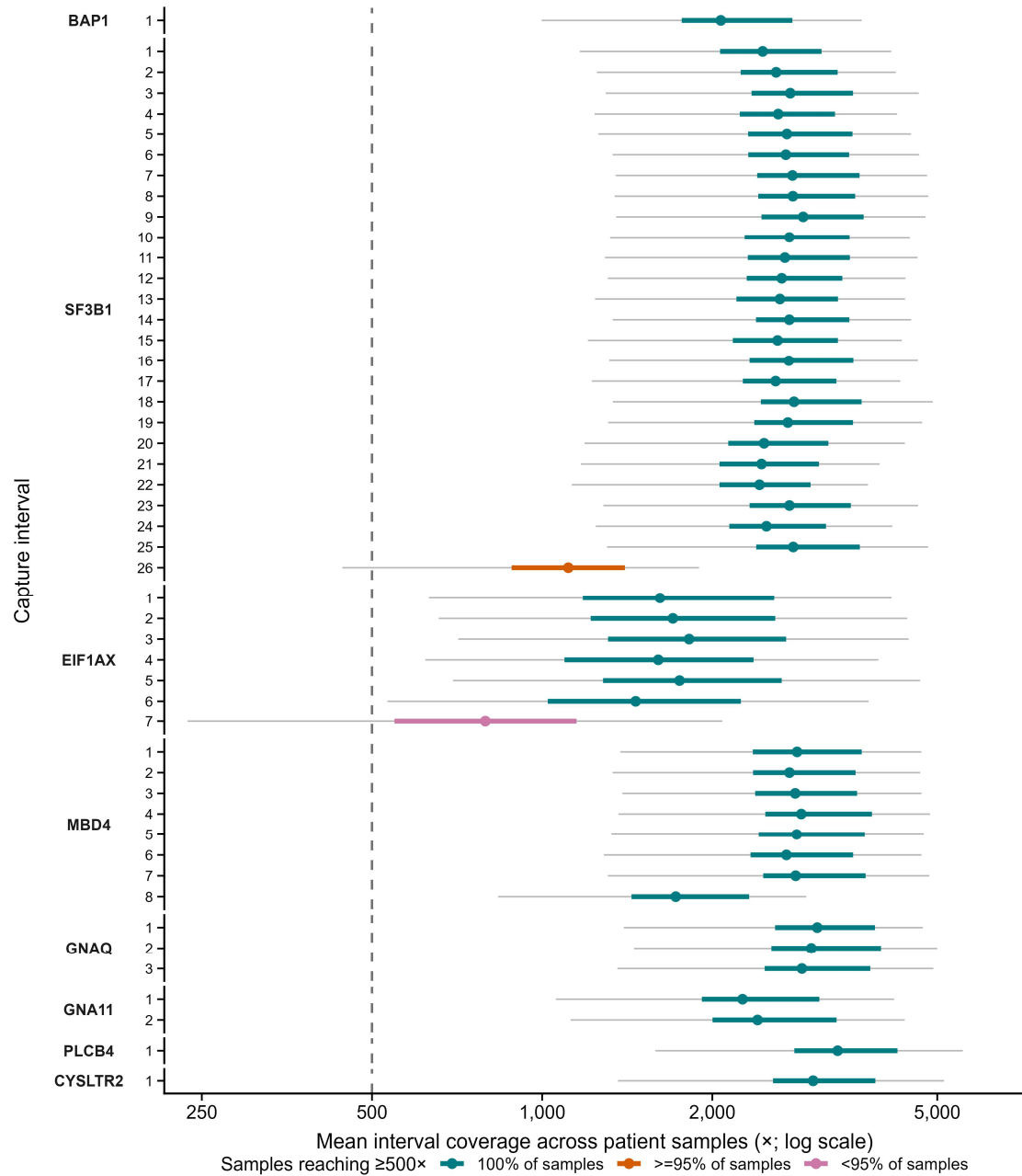
Supplementary Figure S3. Duplex UMI-family-size distribution. Bars show the median percentage of duplex families in each strand-family-size category across 104 patient plasma samples; whiskers show IQRs and labels show median percentages. Categories are based on BA, the smaller strand family; the corresponding AB family may contain additional reads.



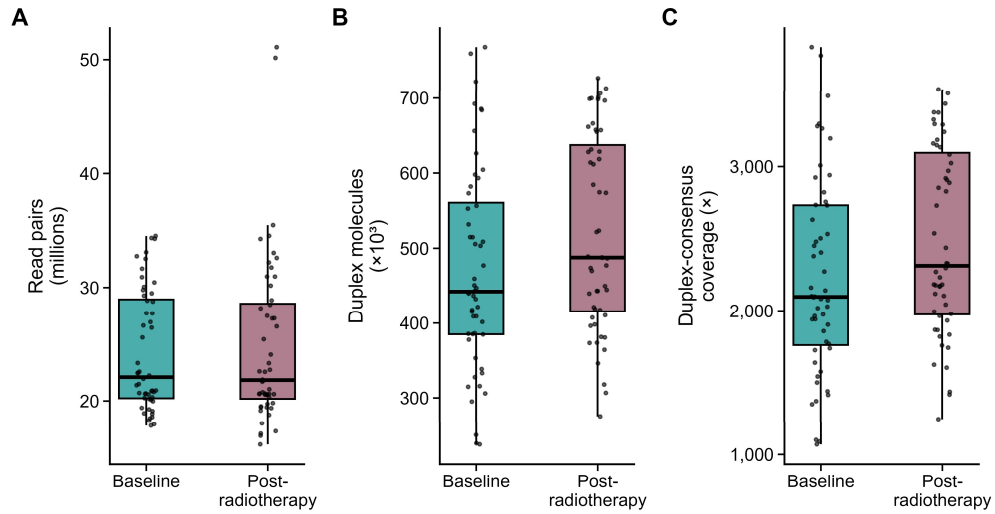
Supplementary Figure S4. Gene- and base-level coverage by sample group. (A) Duplex-consensus coverage across the eight targeted genes in six healthy-control and 104 patient plasma samples. Boxes show medians and IQRs, and points show individual samples. (B) Median percentage of targeted bases covered across increasing coverage thresholds; shaded regions show IQRs.



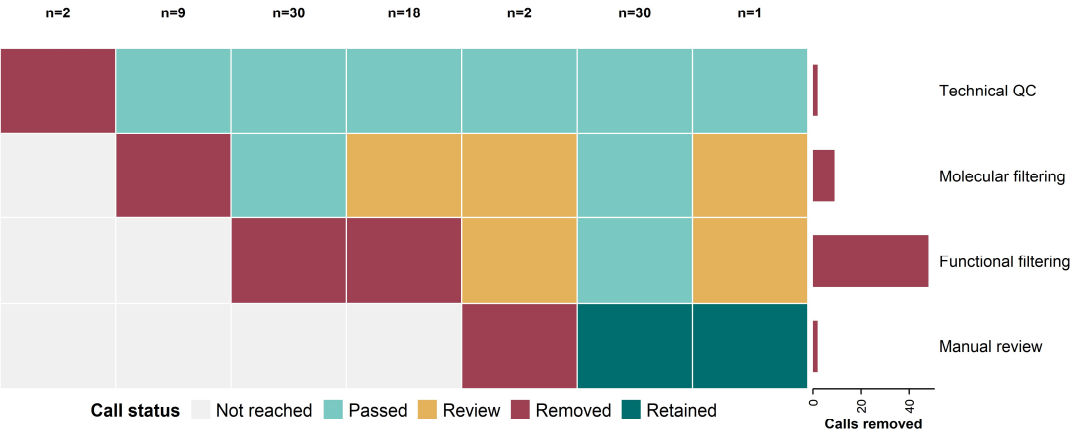
Supplementary Figure S5. Target-interval coverage in patient plasma samples. Coverage of 49 capture intervals across 104 patient plasma samples, grouped by gene. Points show median coverage, thick lines show IQRs, and thin lines show ranges. The dashed line indicates 500 $\times$ coverage. Colours indicate intervals reaching mean coverage $\geq 500\times$ in 100% of samples (teal), at least 95% but less than 100% of samples (orange), or less than 95% of samples (pink).



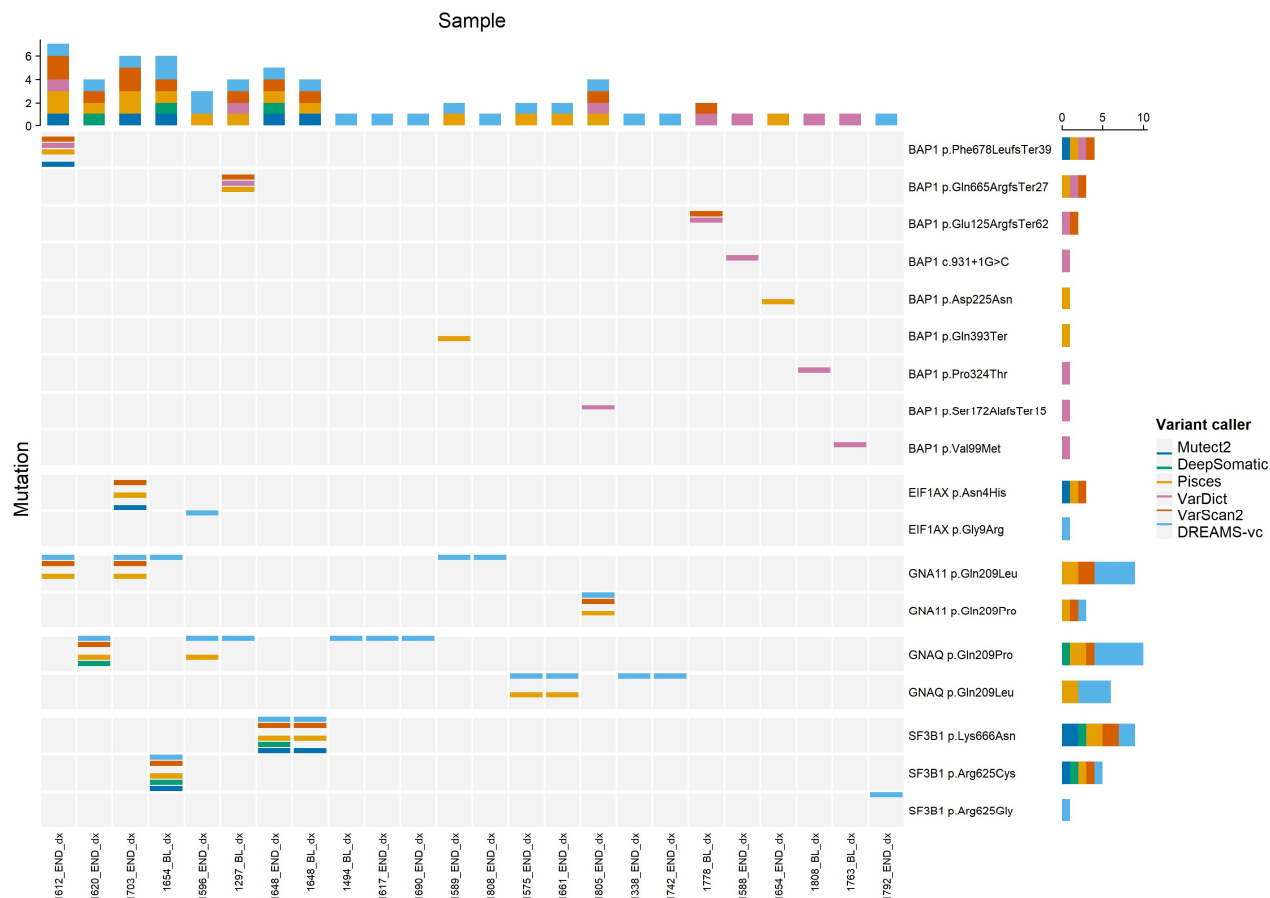
Supplementary Figure S6. Sequencing quality-control measurements by collection timepoint. (A) Sequencing read pairs. (B) Independent duplex molecules. (C) Mean duplex-consensus target coverage. Baseline and post-radiotherapy groups each contain 52 samples from the same patients. Boxes show medians and IQRs, and points show individual samples.



Supplementary Figure S7. Filtering trajectories for candidate variant calls. The heatmap shows how 92 calls moved through technical QC, molecular filtering, functional filtering and manual review. Each column represents one filtering pathway, with the number of calls shown above it. Colours indicate whether calls passed, required review, were removed or were retained. Bars on the right show how many calls were removed at each stage. Overall, 31 calls were retained, with most exclusions occurring during functional filtering.



Supplementary Figure S8. Variant-caller support for retained calls. Rows show the 18 retained genomic variants and columns show the 24 ctDNA-positive plasma samples. Coloured segments identify the variant callers supporting each call; multiple colours indicate support from more than one caller. The top and right bars summarize caller-support observations by sample and variant, respectively.



Supplementary Table S1. Gene-specific filtering criteria. The table summarizes the rules used to pass, review or exclude candidate variants. These include the BAP1 recurrence, molecule-support and functional-evidence filters, and the SF3B1/EIF1AX support filter. Technical QC criteria are described separately in the Methods.

Filter	Exact implemented criterion	Result
BAP1 recurrence filter	A BAP1 variant observed in ≥ 3 patients is excluded only when supported by exactly one caller and one alternate molecule. Other recurrent BAP1 combinations proceed to later assessment.	Exclude recurrent weak calls; otherwise retain or review.
SF3B1/EIF1AX support filter	A single-molecule SF3B1 or EIF1AX call requires median AD ≥ 5 and median BD ≥ 5 . Calls with ≥ 2 alternate molecules do not receive this additional threshold.	Pass or exclude.
BAP1 molecule-support filter	BAP1 calls with ≥ 2 alternate molecules pass. A single-molecule call passes when AD and BD are both ≥ 10 , enters review when both are ≥ 5 but either is < 10 , and otherwise fails.	Pass, review or exclude.
BAP1 functional-evidence filter	Retain technically eligible SGE-confirmed loss-of-function calls or, when no SGE match is available, frameshift, stop-gained, splice-acceptor or splice-donor calls. Recurrent LOF and technically low-support LOF calls enter review. SGE-unchanged, SGE-enriched and calls without LOF evidence are excluded.	Retain, review or exclude.

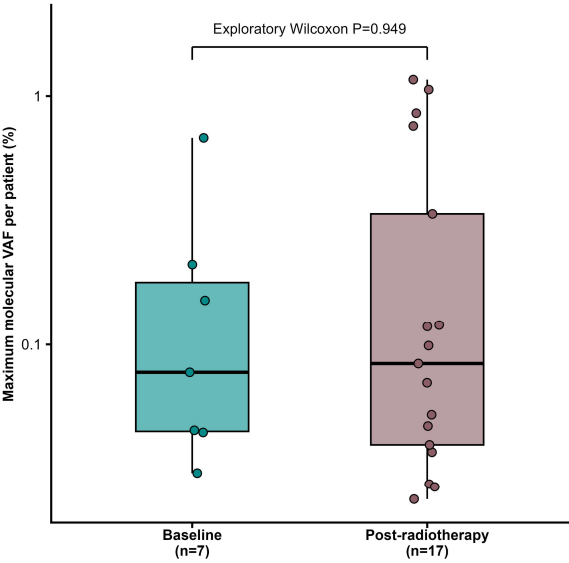
AD and BD denote $qc_median_ad_variant$ and $qc_median_bd_variant$. Technical QC checks and the non-sequential BAP1 sensitivity analysis are reported separately and are not included here.

Supplementary Table S2. Final retained variant calls. The table lists the 31 retained patient–timepoint–variant calls, representing 18 unique genomic variants. Each row reports the patient, timepoint, gene, protein or coding change, genomic variant, consequence, molecular VAF, number of supporting alternate molecules, number of supporting callers and whether retention was automatic or based on manual review.

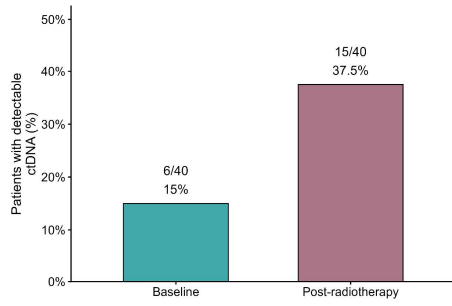
Patient	Endpoint	Variant	Genomic variant	Consequence	VAF (%)	ALT molecules	Callers (n)	Retention
GNAQ								
1297	Baseline	p.Gln209Pro	9_77794572_T_G	Missense	0.064	3	1	Automatic
1338	Post-RT	p.Gln209Leu	9_77794572_T_A	Missense	0.037	1	1	Automatic
1494	Baseline	p.Gln209Pro	9_77794572_T_G	Missense	0.030	1	1	Automatic
1575	Post-RT	p.Gln209Leu	9_77794572_T_A	Missense	0.120	5	2	Automatic
1596	Post-RT	p.Gln209Pro	9_77794572_T_G	Missense	0.077	2	2	Automatic
1617	Post-RT	p.Gln209Pro	9_77794572_T_G	Missense	0.024	1	1	Automatic
1620	Post-RT	p.Gln209Pro	9_77794572_T_G	Missense	0.854	33	4	Automatic
1661	Post-RT	p.Gln209Leu	9_77794572_T_A	Missense	0.084	3	2	Automatic
1690	Post-RT	p.Gln209Pro	9_77794572_T_G	Missense	0.047	2	1	Automatic
1742	Post-RT	p.Gln209Leu	9_77794572_T_A	Missense	0.040	1	1	Automatic
GNA11								
1589	Post-RT	p.Gln209Leu	19_3118944_A_T	Missense	0.075	1	1	Automatic
1612	Post-RT	p.Gln209Leu	19_3118944_A_T	Missense	0.614	12	3	Automatic
1654	Baseline	p.Gln209Leu	19_3118944_A_T	Missense	0.056	2	1	Automatic
1703	Post-RT	p.Gln209Leu	19_3118944_A_T	Missense	0.347	13	3	Automatic
1805	Post-RT	p.Gln209Pro	19_3118944_A_C	Missense	0.099	3	3	Automatic
1808	Post-RT	p.Gln209Leu	19_3118944_A_T	Missense	0.027	1	1	Automatic
BAP1								
1297	Baseline	p.Gln665ArgfsTer27	3_52402664_TG_T	Frameshift	0.151	7	3	Automatic
1588	Post-RT	c.931+1G>C	3_52405764_C_G	Splice donor	0.052	2	1	Automatic
1589	Post-RT	p.Gln393Ter	3_52404526_G_A	Stop gained	0.335	5	1	Automatic
1612	Post-RT	p.Phe678LeufsTer39	3_52402624_A_AC	Frameshift	1.164	25	4	Automatic
1654	Post-RT	p.Asp225Asn	3_52406363_C_T	Missense	0.070	2	1	Manual
1763	Baseline	p.Val99Met	3_52408038_C_T	Missense	0.077	2	1	Automatic
1778	Baseline	p.Glu125ArgfsTer62	3_52407960_CA_C	Frameshift	0.045	1	2	Automatic
1805	Post-RT	p.Ser172AlafsTer15	3_52407239_CT_C	Frameshift	0.033	1	1	Manual
1808	Baseline	p.Pro324Thr	3_52405256_G_T	Missense	0.044	1	1	Automatic
SF3B1								
1648	Baseline	p.Lys666Asn	2_197402635_C_A	Missense	0.681	12	4	Automatic
1648	Post-RT	p.Lys666Asn	2_197402635_C_A	Missense	0.759	13	5	Automatic
1654	Baseline	p.Arg625Cys	2_197402760_G_A	Missense	0.210	9	5	Manual
1792	Post-RT	p.Arg625Gly	2_197402760_G_C	Missense	0.028	1	1	Automatic
EIF1AX								
1596	Post-RT	p.Gly9Arg	X_20138614_C_G	Missense	0.118	1	1	Automatic
1703	Post-RT	p.Asn4His	X_20141631_T_G	Missense	1.062	10	3	Manual

One row represents one retained patient-timepoint-variant call. Molecular VAF is calculated from duplex-consensus reads. Detailed caller identities are provided in Supplementary Figure S8.

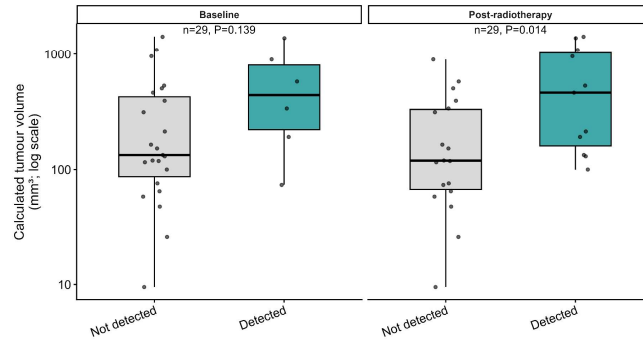
Supplementary Figure S7. Maximum molecular VAF among ctDNA-positive samples before and after radiotherapy. Distribution of maximum molecular variant allele fraction (VAF) among samples with detectable circulating tumour DNA (ctDNA) at baseline and post-radiotherapy. Maximum molecular VAF represents the highest molecular VAF among retained variants within each sample. Only ctDNA-positive samples are included.



A

ctDNA detectabilityClinical cohort; exact paired $P=0.049$ 

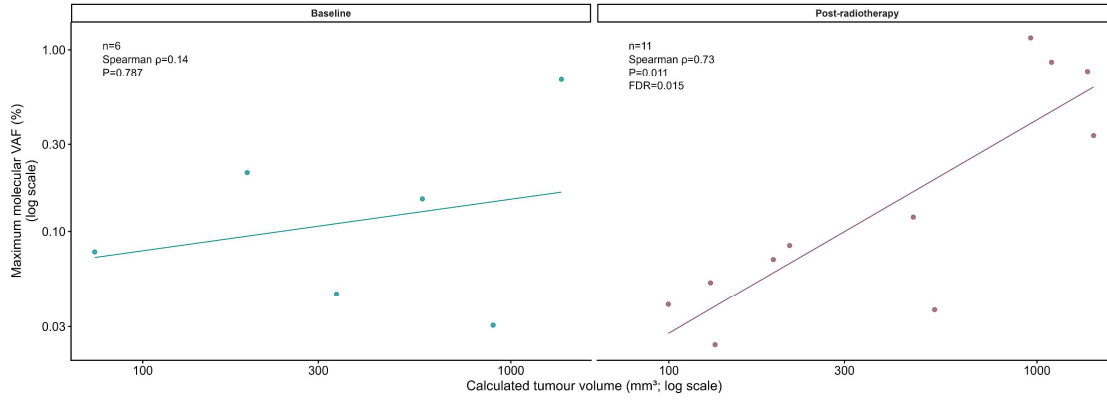
B

Calculated tumour volume by ctDNA detectability

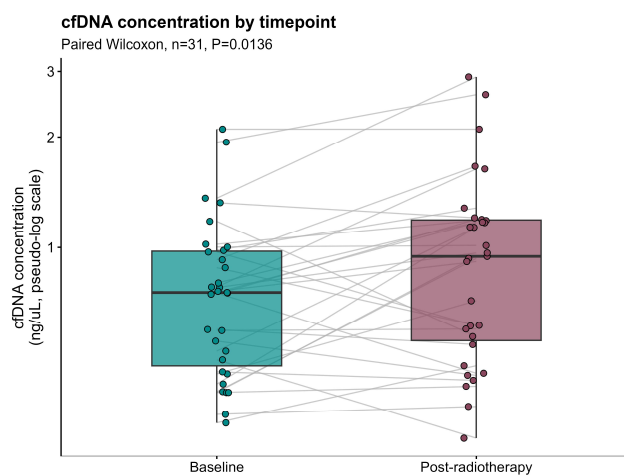
C

Calculated tumour volume and maximum molecular VAF

ctDNA-positive samples only; fitted lines are descriptive



Supplementary Figure S11. Total cfDNA concentration before and after radiotherapy. Paired total cell-free DNA (cfDNA) concentrations measured at baseline and post-radiotherapy in 31 patients. Points represent individual measurements, and lines connect paired samples from the same patient. Boxes show the median and interquartile range. The y-axis uses a pseudo-logarithmic scale. The P value was calculated using a paired Wilcoxon signed-rank test. Total cfDNA was quantified within the cfDNA-sized fraction and does not specifically represent tumour-derived ctDNA.



Supplementary Table S3. Clinical characteristics according to ctDNA detectability. Baseline clinical characteristics stratified by ctDNA detection status at baseline and post-radiotherapy. Continuous variables are presented as median (interquartile range) [available n] and were compared using Wilcoxon rank-sum tests. Categorical variables are presented as n/N (%) and were compared using Fisher's exact tests. Benjamini–Hochberg false discovery rate (FDR) correction was applied separately to the three tumour-size comparisons within each timepoint. Missing values were not imputed.

Supplementary Table S3. Clinical variables by ctDNA detectability

Clinical subset, N=40; ctDNA detected in 6/40 (15.0%) at Baseline and 15/40 (37.5%) post-radiotherapy; exact paired P=0.049

Characteristic	Baseline				Post-radiotherapy			
	ctDNA– N=34	ctDNA+ N=6	P	FDR	ctDNA– N=25	ctDNA+ N=15	P	FDR
Age, years	67.5 (56.0–73.0) [34]	70.0 (68.0–73.5) [6]	0.483	—	67.0 (53.0–72.0) [25]	69.0 (62.5–76.0) [15]	0.057	—
Sex: male	23/34 (67.6%)	3/6 (50.0%)	0.646	—	15/25 (60.0%)	11/15 (73.3%)	0.502	—
Affected eye: left	20/33 (60.6%)	3/6 (50.0%)	0.674	—	16/25 (64.0%)	7/14 (50.0%)	0.503	—
Apical height, mm	3.5 (2.8–5.8) [23]	6.9 (3.9–7.6) [6]	0.257	0.257	3.1 (2.3–5.2) [18]	6.5 (3.5–9.0) [11]	0.024	0.037
Largest basal diameter, mm	10.0 (8.6–12.5) [23]	13.1 (10.4–16.0) [6]	0.195	0.257	9.9 (8.1–11.8) [18]	12.0 (10.4–15.6) [11]	0.045	0.045
Calculated tumour volume, mm ³	133.5 (87.7–426.3) [23]	455.5 (228.2–815.2) [6]	0.139	0.257	119.5 (67.6–330.4) [18]	461.4 (162.9–1028.4) [11]	0.014	0.037
Tissue risk: high	11/16 (68.8%)	3/3 (100.0%)	0.530	—	10/14 (71.4%)	4/5 (80.0%)	1.000	—

Continuous variables are median (IQR) [available n]; categorical variables are n/N (%) for the stated level. Wilcoxon rank-sum tests were used for continuous variables and Fisher exact tests for categorical variables.

FDR values use Benjamini–Hochberg correction across the three tumour-size comparisons within each timepoint; FDR is not shown for the other exploratory clinical comparisons. Clinical variables are baseline/static covariates; ctDNA status is timepoint-specific. Missing values were not imputed.

Supplementary Table S4. Tumour measurements and maximum molecular VAF according to ctDNA detectability and timepoint. Tumour measurements stratified by ctDNA detection status, maximum molecular variant allele fraction (VAF) among ctDNA-positive samples, and Spearman correlations between maximum molecular VAF and clinical variables. Continuous measurements are presented as median (interquartile range) [available n]. Benjamini–Hochberg false discovery rate (FDR) correction was applied separately to the three tumour-size comparisons and to the four correlation analysis in each point. Minimum observed values are descriptive

Tumour measurements and maximum molecular VAF by ctDNA detectability and timepoint								
Measure or analysis	Baseline				Post-radiotherapy			
	ctDNA–	ctDNA+	P	FDR	ctDNA–	ctDNA+	P	FDR
Tumour size by ctDNA detectability								
Apical height (mm)	3.5 (2.8–5.8) [23]	6.9 (3.9–7.6) [6]	0.257	0.257	3.1 (2.3–5.2) [18]	6.5 (3.5–9.0) [11]	0.024	0.037
Largest basal diameter (mm)	10.0 (8.6–12.5) [23]	13.1 (10.4–16.0) [6]	0.195	0.257	9.9 (8.1–11.8) [18]	12.0 (10.4–15.6) [11]	0.045	0.045
Calculated tumour volume (mm ³)	133.5 (87.7–426.3) [23]	455.5 (228.2–815.2) [6]	0.139	0.257	119.5 (67.6–330.4) [18]	461.4 (162.9–1028.4) [11]	0.014	0.037
Maximum molecular VAF among ctDNA-positive samples								
Median maximum molecular VAF, %	—	0.114% (0.053–0.195) [6]	—	—	—	0.084% (0.043–0.547) [15]	—	—
Maximum molecular VAF and clinical features								
Age	—	p=0.348 [6]	0.499	0.872	—	p=0.115 [15]	0.683	0.683
Apical height	—	p=0.086 [6]	0.872	0.872	—	p=0.784 [11]	0.004	0.015
Largest basal diameter	—	p=0.314 [6]	0.544	0.872	—	p=0.738 [11]	0.010	0.015
Calculated tumour volume (mm ³)	—	p=0.143 [6]	0.787	0.872	—	p=0.727 [11]	0.011	0.015
Minimum observed values among ctDNA-positive samples								
Minimum observed apical height (mm)	—	2.6 [6]	—	—	—	3.0 [11]	—	—
Minimum observed largest basal diameter (mm)	—	8.0 [6]	—	—	—	8.0 [11]	—	—
Minimum observed calculated tumour volume (mm ³)	—	74.1 [6]	—	—	—	99.7 [11]	—	—
Minimum observed maximum molecular VAF	—	0.030% [6]	—	—	—	0.024% [15]	—	—
Tumour measurements and maximum molecular VAF are median (IQR) [available n]. Spearman correlations are shown as p [available n] and include ctDNA-positive samples only. P values for tumour measurements compare ctDNA-negative and ctDNA-positive patients using Wilcoxon rank-sum tests.								
FDR correction was applied separately within each timepoint across the three tumour-size comparisons or the four maximum-VAF correlations. Minimum observed values are descriptive observations and do not represent analytical or biological detection thresholds.								

Supplementary Figure S10. Changes in cfDNA fragment patterns after radiotherapy.

(A) The overall fragment-size distributions were similar at baseline and post-radiotherapy. (B) Median within-patient changes differed across the fitted fragment-size components. (C–D) No clear changes were observed in the overall amplitude ratio or entropy ratio. (E) Entropy changes are shown for grouped fragment sizes. (F) The fitted amplitude of the 121-bp component decreased after radiotherapy (BH-adjusted $P=0.007$). (G) The percentage of fragments measuring 115–125 bp also decreased ($P=0.003$); this was an exploratory supporting analysis. Analyses included 52 paired patients. Lines connect samples from the same patient, boxes show medians and interquartile ranges, and dashed lines indicate no change. P values were calculated using paired Wilcoxon signed-rank tests.

